

Cognitive Load Mapper

Stats 418 Final Presentation - June 2, 2026

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<https://cogload-frontend-98299138336.us-west2.run.app>

Overview

Problem Statement

How can cognitive load across an article be modeled using readability, lexical, and sentiment features?

Motivation

Observing how cognitive load fluctuates throughout a long-form text provides readers with another way to understand the difficulty of a piece and provides writers/editors with paragraph-level feedback for revision.

Key Objectives

Build a complexity scoring pipeline and classify articles into data-driven structural shapes via modeling.

Deliver an interactive Streamlit app where users browse articles, visualize complexity trajectories, and request LLM-assisted rewrites toward a target gradient shape.

Data Collection

Publication exploration & selection

6 Substack publications (public HTML)



Scraping

(requests + BeautifulSoup)

Article filtering

(engagement metrics, ≥ 800 words, ≥ 6 paragraphs)



Paragraph segmentation

(split on blank lines)

Final Dataset Characteristics

<i>Publications</i>	6
<i>Articles</i>	1128
<i>Paragraphs</i>	30527

Article Statistics

<i>Mean word count</i>	1960
<i>Max word count</i>	11530

Notable challenges:

No engagement metrics for certain publications

- ☐ selection and filtering

Insufficient amount of articles

- ☐ higher number for collection

Long scrape time

- ☐ per-article checkpointing

Modeling

Feature Engineering

Feature Type	Count
Embedding distances	3
Sentiment	7
Trajectory/ shape	4
Readability	4
Core gradient	8

Modeling Approach:

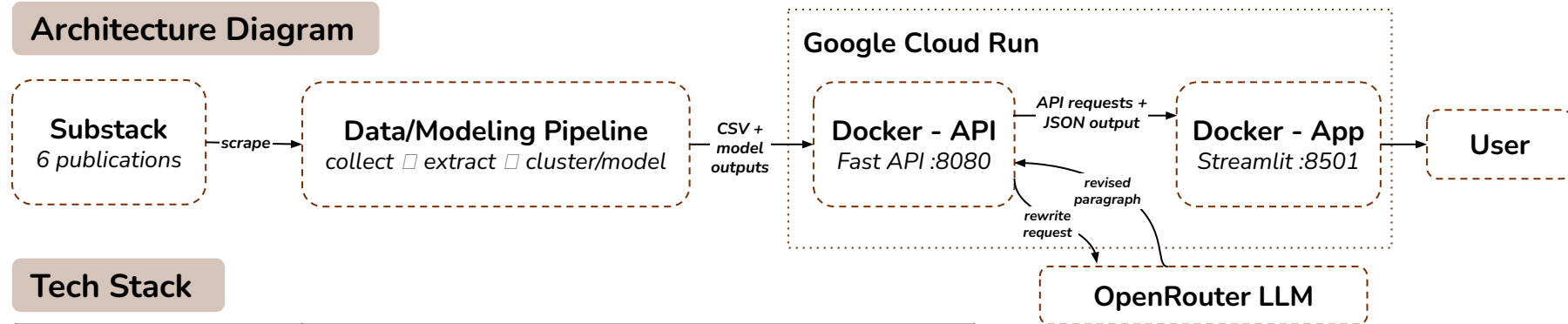
- Feature sets compared (ablation study)
- Regression and classification
- Publication-aware cross-validation



Model	ROC-AUC	MAE	RMSE
Baseline: FK grade only	0.58	0.75	0.99
Baseline: mean complexity only	0.56	0.76	1.00
Readability only (4 features)	0.57	0.76	1.01
Full gradient (25 features)	0.66	0.69	0.92
Gradient + controls (31 features)	0.68	0.68	0.91
Ablation: no shape features	0.67	0.68	0.90
Ablation: no embedding features	0.66	0.6	0.92

Architecture

Architecture Diagram



Tech Stack

Step	Tech
Data collection	Python, requests, BeautifulSoup, lxml
Feature extraction	spaCy, textstat, nltk, scikit-learn, sentence-transformers
Modeling	scikit-learn
API	FastAPI, Pydantic, slowapi
Application	Streamlit, OpenRouter
Deployment	Docker, Google Cloud Run

Application

Interaction

Explore

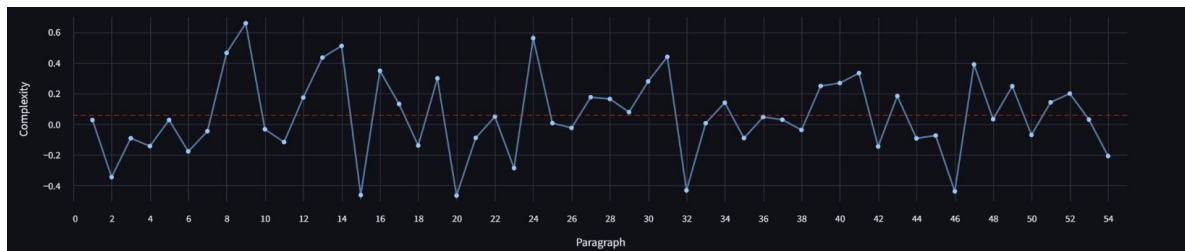
Pick publication, filter by shape, choose article

Diagnose

Read metrics and inspect paragraph-to-paragraph cognitive load trajectory

Rewrite

Flag up to 3 paragraphs and compare original vs LLM rewrite



Target shape
ramp

Resolution → Ramp — Complexity builds toward the end

3 paragraphs will be targeted for rewriting.

Rewrite flagged paragraphs

3 paragraph(s) rewritten.

\$9 — Too complex at position 0.15 — early paragraphs should build gradually toward a ramp.

Original	Rewritten
Most attempts at construction innovation have a very long history. Construction startup Katera's attempt to achieve low costs through factory-built, mass-produced homes was predated by things like the Lustron Home in the 1940s, Operation Breakthrough in the 1960s, and Toyota's homebuilding division . Mechanical bricklayers have been attempted repeatedly since the late 19th century .	Most construction innovations have a long history. Katera's effort to lower costs by building factory-made, mass-produced homes followed earlier projects such as the Lustron Home in the 1940s, Operation Breakthrough in the 1960s, and Toyota's homebuilding division. People have also tried using mechanical bricklayers repeatedly since the late 19th century.

Browse Articles

Publication

- ☒ Construction Physics
- ☐ Experimental History
- ☐ One Useful Thing
- ☐ The Intrinsic Perspective
- ☐ Age of Invention

Filter by shape

(any) ▾

> What do shapes mean?

Select article

50 Things I've Learned Writing C... ▾

Cognitive Load Mapper

Visualize how cognitive load flows through an article.

50 Things I've Learned Writing Construction Physics

www.construction-physics.com · 2025-04-17 · [Read original](#)

Gradient Shape

resolution

Word Count

1937

Mean Complexity

0.06

Engagement Score ⓘ

0.91

Complexity fades toward the end

> Full Article Text

Complexity Trajectory



Red dashed line = mean complexity

Paragraphs

>  \$1 · complexity 0.03 · ↓ below avg

>  \$2 · complexity -0.34 · ↓ below avg

>  \$3 · complexity -0.09 · ↓ below avg

>  \$4 · complexity -0.14 · ↓ below avg

>  \$5 · complexity 0.03 · ↓ below avg

>  \$6 · complexity -0.18 · ↓ below avg

>  \$7 · complexity -0.05 · ↓ below avg

>  \$8 · complexity 0.46 · ↑ above avg

>  \$9 · complexity 0.66 · ↑ above avg

>  \$10 · complexity -0.03 · ↓ below avg

>  \$11 · complexity -0.12 · ↓ below avg

>  \$12 · complexity 0.17 · ↑ above avg

>  \$13 · complexity 0.43 · ↑ above avg

Target shape

ramp



Resolution → Ramp — Complexity builds toward the end

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Rewrite flagged paragraphs

3 paragraph(s) rewritten.

\$9 — *Too complex at position 0.15 — early paragraphs should build gradually toward a ramp.*

Original

Most attempts at construction innovation have a very long history. Construction startup Katerra's attempt to achieve low costs through factory-built, mass-produced homes was predated by things like the Lustron Home in the 1940s, Operation Breakthrough in the 1960s, and Toyota's homebuilding division . Mechanical bricklayers have been attempted repeatedly since the late 19th century .

Rewritten

Most construction innovations have a long history. Katerra's effort to lower costs by building factory-made, mass-produced homes followed earlier projects such as the Lustron Home in the 1940s, Operation Breakthrough in the 1960s, and Toyota's homebuilding division. People have also tried using mechanical bricklayers repeatedly since the late 19th century.

\$46 — *Too simple at position 0.85 — late paragraphs should carry the most complexity in a ramp.*

Original

Bell Labs has done more Nobel Prize-winning work than the next six private corporations combined . Most of its prize-winning work was done by researchers hired after the company invented the transistor, at the peak of its prestige.

Rewritten

Bell Laboratories has accumulated a greater number of Nobel Prize-winning achievements than the aggregate total of the six next-largest private corporations, a testament to its unparalleled research ecosystem. The lion's share of these laureate-caliber contributions emerged from scientists who joined the institution after its seminal invention of the transistor in 1947, a period when Bell Labs enjoyed its zenith of scientific influence, institutional resources, and cultural prestige.

Conclusion

Technical Challenges

Engagement metrics inconsistently structured, weak modelling, two separate containers (app + API)

Role of AI Assistants

Initial project design, model iteration, debugging, LLM implementation in application, Streamlit UX improvements

Key Lessons

Engagement information is noisy, within-publication normalization is essential, thinking about user interaction on application is important

Future Work

Modeling improvements, expansion to more publications, expansion to other platforms, URL import in the app, LLM as a rater